Chapter 1: Risk in Perspective

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Section 1: Introduction of Risks and Risk Management

What is Risk?

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"hazard, a chance of bad consequences, loss or exposure to mischance"

"any event or action that may adversely affect an organization's ability to achieve its objectives and execute its strategies"

"the quantifiable likelihood of loss or less-than-expected returns"

Financial Risks

We concentrate on the following sources of risk.

- Market Risk: risk associated with fluctuations in value of traded assets.
- Credit Risk: risk associated with uncertainty that debtors will honor their financial obligations.
- Operational Risk: risk associated with possibility of human error, IT failure, dishonesty, natural disaster etc.

This is a non-exhaustive list; other sources of risk such as liquidity risk possible.

Why Manage Financial Risks?

- A societal view

  The danger that problems in a single financial institution may spill over and disrupt the normal functioning of the entire financial system.

- The shareholder's view

  Proper RM can increase the value of a corporation and shareholder value.

  - When does RM increase the value of a firm, and which risks should be managed?
  - How should RM concerns factor into investment policy and capital budgeting?

- Economic capital

Rationales for Risk Management

- RM can reduce tax costs. Under a typical tax regime the amount of tax to be paid by a corporation is a *convex* function of its profits; by reducing the variability in a firm's cash flow, RM can therefore lead to a higher expected after-tax profit.
- RM can be beneficial, since a company may (and usually will) have better access to capital markets than individual investors.
- RM can increase the firm value in the presence of bankruptcy costs, as it makes bankruptcy less likely.
- RM can reduce the impact of costly external financing on the firm value, as it facilitates the achievement of optimal investment.

Insurance Risks

The insurance industry also has a longstanding relationship with risk. The Institute and Faculty of Actuaries use the following definition of the actuarial profession:

Actuaries are respected professionals whose innovative approach to making business successful is matched by a responsibility to the public interest. Actuaries identify solutions to financial problems. They manage assets and liabilities by analyzing past events, assessing the present risk involved and modeling what could happen in the future.

An additional risk category entering through insurance is underwriting risk: the risk inherent in insurance policies sold.

A general view of insurer risks

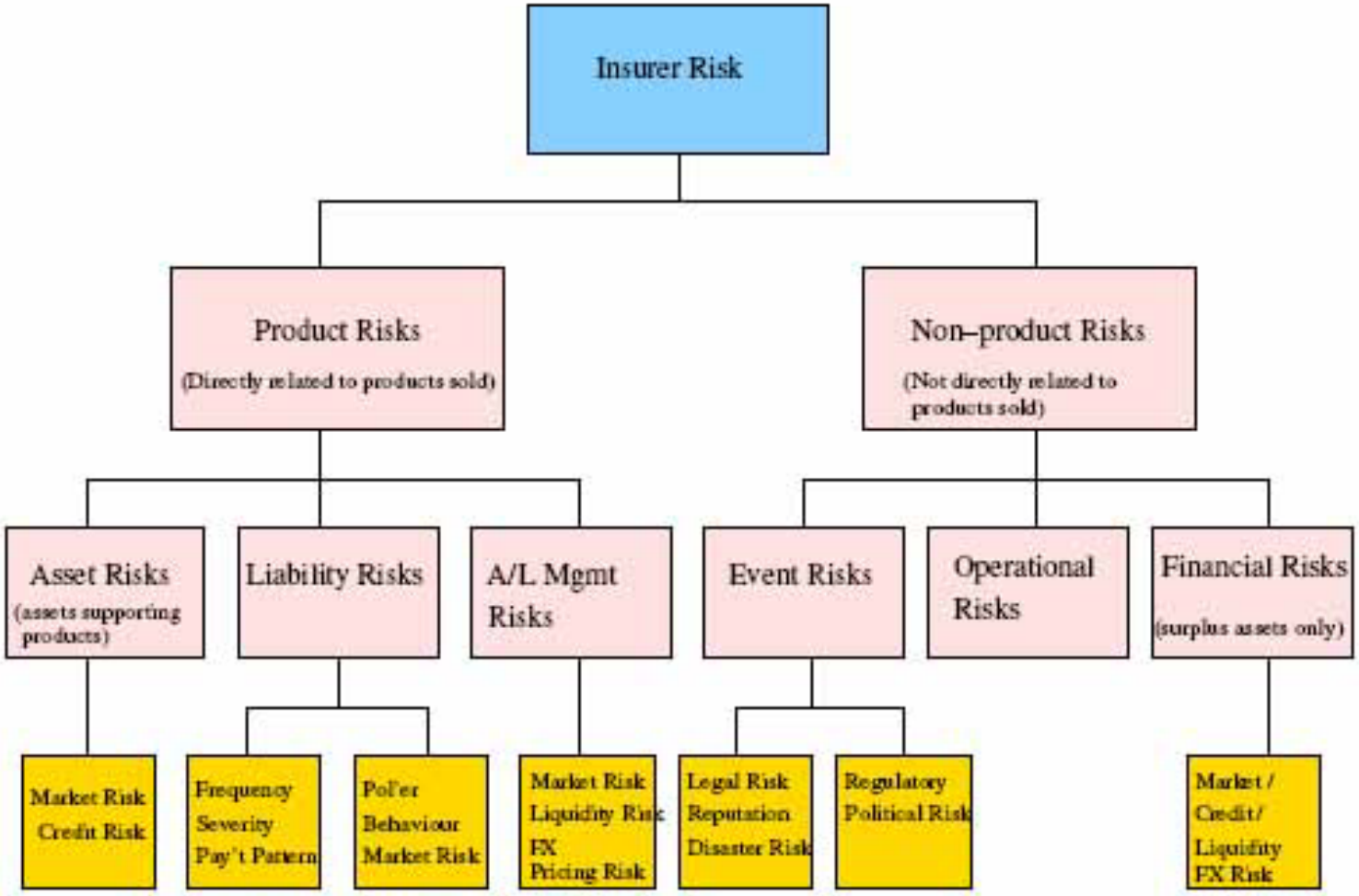

Figure 1.1

Overall management of risk by an insurer

- Ingredients:
  - design, pricing and underwriting of its insurance policies
  - selection of assets backing the policies
  - estimation of the size and volatility of the liabilities associated with those policies
  - determination of the insurer's capital needs
  - claims management
  - updating of all these elements over time as more data and other information become available or because the underlying risk processes change
  - adequate/sound disclosure/communication process to key stakeholders, e.g. management, investors, policyholders and regulators
  - periodic financial condition analysis, providing a prospective view of the company as a whole.

The actuarial control cycle

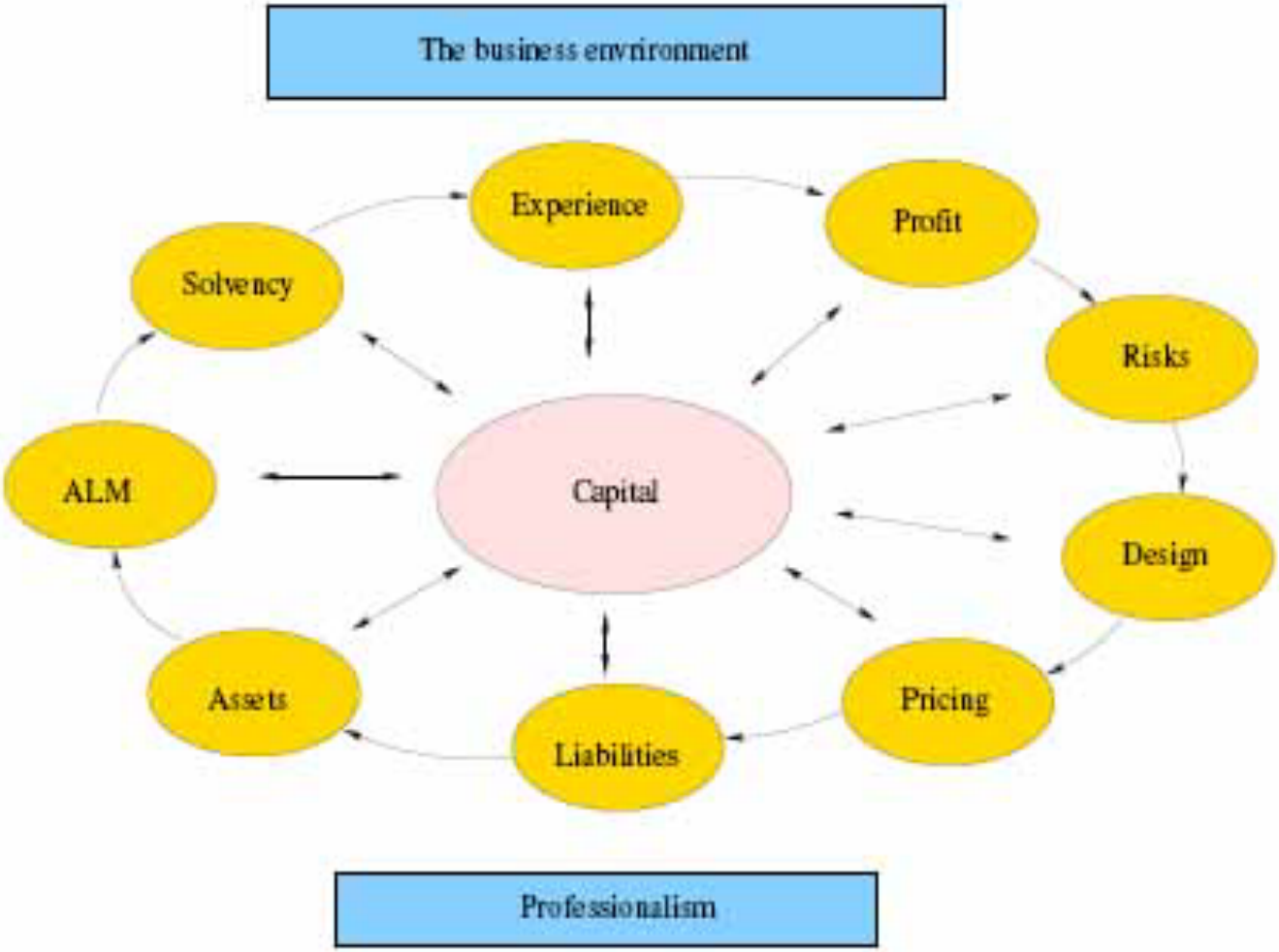

Figure 1.2

Overall management of risk by an insurer

- Multitude of actuarial tools enter
  - collective risk model
  - aggregate risk models
  - diffusion models and other stochastic processes
  - multi-state models
  - cash flow models
- Specific concerns are:
  - time horizon (long-term RM)
  - combining risks (copulae)
  - appropriate risk measures (coherence)
  - extreme event risk (EVT)
  - and other . . .

Key components of the new regulatory framework

- Summary:
  - principle-based solvency requirement
  - risk-based
  - expected shortfall as risk measure (beyond VaR!)
  - modeling of dependencies through "increased correlations" or copula based approach, stress scenarios (EVT)
  - transition from conservative standard-factor-model to a full internal model Ansatz (allowing for intermediate levels)
  - potential issues: time-horizon (life versus non-life);confidence level (per risk category, overall level)

Modeling Financial Risks

To model risk we use language of probability theory. Risks are represented by random variables mapping unforeseen future states of the world into values representing profits and losses.

The risks which interest us are aggregate risks. In general we consider a portfolio which might be

- a collection of stocks and bonds;
- a book of derivatives;
- a collection of risky loans;
- a financial institution's overall position in risky assets.

Portfolio Values and Losses

Consider a portfolio and let V_t denote its value at time t; we assume this random variable is observable at time t.

Suppose we look at risk from perspective of time t and we consider the time period $[t, t+1]$. The value V_{t+1} at the end of the time period is unknown to us.

The distribution of $(V_{t+1} - V_t)$ is known as the profit-and-loss or P&L distribution. We denote the loss by $L_{t+1} = -(V_{t+1} - V_t)$. By this convention, losses will be positive numbers and profits negative.

We refer to the distribution of L_{t+1} as the loss distribution.

Introducing Risk Factors

The Value V_t of the portfolio/position will be modeled as a function of time and a set of d underlying risk factors. We write

期权、期货的价值就受到时间因素的影响

$$V_t = f(t, \mathbf{Z}_t) \tag{1}$$

where $\mathbf{Z}_t = (Z_{t,1}, \cdots, Z_{t,d})'$ is the risk factor vector. This representation of portfolio value is known as a mapping.

Examples of typical risk factors:

- (logarithmic) prices of financial assets
- yields
- (logarithmic) exchange rates

Risk Factor Changes

We define the time series of risk factor changes by

$$X_t := \mathbf{Z}_t - \mathbf{Z}_{t-1}$$

Typically, historical risk factor time series are available and it is of interest to relate the changes in these underlying risk factors to the changes in portfolio value. We have

$$\begin{aligned} L_{t+1} &= -(V_{t+1} - V_t) \\ &= -(f(t+1, \mathbf{Z}_{t+1}) - f(t, \mathbf{Z}_t)) \\ &= -(f(t+1, \mathbf{Z}_t + \mathbf{X}_{t+1}) - f(t, \mathbf{Z}_t)) \end{aligned} \tag{2}$$

不确定项

The Loss Operator

Since the risk factor values $\mathbf{Z}_t$ are known at time t the loss L_{t+1} is determined by the risk factor changes $\mathbf{X}_{t+1}$.

Given realization $\mathbf{z}_t$ of $\mathbf{Z}_t$, the loss operator at time t is defined as

$$l_{[t]}(\mathbf{x}) := -(f(t+1, \mathbf{z}_t + \mathbf{x}) - f(t, \mathbf{z}_t)) \tag{3}$$

so that

$$L_{t+1} = l_{[t]}(\mathbf{X}_{t+1})$$

From the perspective of time t the loss distribution of L_{t+1} is determined by the multivariate distribution of $\mathbf{X}_{t+1}$.

But which distribution exactly? Conditional distribution of L_{t+1} given history up to and including time t or unconditional distribution under assumption that (X_t) form stationary time series?

Conditional or Unconditional Loss Distribution?

This issue is related to the time series properties of $(\mathbf{X}_t)_{t\in\mathbb{N}}$, the series of risk factor changes. If we assume that $\mathbf{X}_t, \mathbf{X}_{t-1}, \cdots$, are iid random vectors, the issue does not arise. But, if we assume that they form a strictly stationary multivariate time series then we must differentiate between conditional and unconditional.

Many standard accounts of risk management fail to make the distinction between the two.

If we cannot assume that risk factor changes form a stationary time series for at least some window of time extending from the present back into intermediate past, then any statistical analysis of loss distribution is difficult.

The Conditional Problem

Let $\mathscr{F}_t$ represent the history of the risk factors up to the present.

各种风险因子不同时间的组合

More formally $\mathscr{F}_t$ is sigma algebra generated by past and present risk factor changes $(\mathbf{X}_s)s \leq t$.

In the conditional problem we are interested in the distribution of $L_{t+1} = l_{[t]}(\mathbf{X}_{t+1})$ given $\mathscr{F}_t$, i.e. the conditional (or predictive) loss distribution for the next time interval given the history of risk factor developments up to present.

This problem forces us to model the dynamics of the risk factor time series and to be concerned in particular with predicting volatility. This seems the most suitable approach to market risk.

The Unconditional Problem

In the unconditional problem we are interested in the distribution of $L_{t+1} = l_{[t]}(\mathbf{X})$ when $\mathbf{X}$ is a generic vector of risk factor changes with the same distribution $F_{\mathbf{X}}$ as $\mathbf{X}_t, \mathbf{X}_{t-1}, \cdots$.

When we neglect the modeling of dynamics we inevitably take this view. Particularly when the time interval is large, it may make sense to do this. Unconditional approach also typical in credit risk.

More Formally:

Conditional loss distribution: distribution of $l_{[t]}(\cdot)$ under $F_{[X_{t+1}|\mathscr{F}_t]}$.

Unconditional loss distribution: distribution of $l_{[t]}(\cdot)$ under $F_{\mathbf{X}}$.

Example: Portfolio of Stocks

Consider d stocks; let α_i denote number of shares in stock i at time t and let $S_{t,i}$ denote price.

The risk factors: following standard convention we take logarithmic prices as risk factors $Z_{t,i} = \log S_{t,i}$, $1 \leq i \leq d$.

The risk factor changes: in this case these are $X_{t+1,i} = \log S_{t+1,i} - \log S_{t,i}$, which correspond to the so-called log-returns of the stock.

The Mapping (1)

$$V_t = \sum_{i=1}^{d} \alpha_i S_{t,i} = \sum_{i=1}^{d} \alpha_i e^{Z_{t,i}}. \tag{4}$$

Example

The Loss (2)

$$
\begin{aligned}
L_{t+1} &= -\left(\sum_{i=1}^{d} \alpha_i e^{Z_{t+1,i}} - \sum_{i=1}^{d} \alpha_i e^{Z_{t,i}}\right) \\
&= -V_t \sum_{i=1}^{d} \omega_{t,i}\left(e^{X_{t+1,i}} - 1\right) \qquad (5)
\end{aligned}
$$

where $\omega_{t,i} = \alpha_i S_{t,i}/V_t$ is relative weight of stock i at time t.

The loss operator (3)

$$
l_{[t]}(x) = -V_t \sum_{i=1}^{d} \omega_{t,i}(e^{x_i} - 1)
$$

可以一阶近似

Linearization of Loss

Recall the general formula (2) for the loss L_{t+1} in time period $[t, t+1]$. If the mapping f is differentiable we may use the following first order approximation for the loss 线性近似可以保持分布

$$L_{t+1}^{\Delta} = -\left(f_t(t, \mathbf{Z}_t) + \sum_{i=1}^{d} f_{z_i}(t, \mathbf{Z}_t) X_{t+1,i} \right) \tag{6}$$

f_{z_i} is partial derivative of mapping with respect to risk factor i

f_t is partial derivative of mapping with respect to time t.

The term $f_t(t, \mathbf{Z}_t)$ only appears when mapping explicitly features time (derivative portfolios) and is sometimes neglected.

Linearized Loss Operator

Recall the loss operator (3) which applies at time t.

We can obviously also define a linearized loss operator

$$l^{\Delta}_{[t]}(\mathbf{x}) = -\left(f_t(t, \mathbf{z}_t) + \sum_{i=1}^{d} f_{z_i}(t, \mathbf{z}_t) x_i\right) \tag{7}$$

where notation is as in previous slide and $\mathbf{z}_t$ is realization of $\mathbf{Z}_t$.

Linearization is convenient because linear functions of the risk factor changes may be easier to handle analytically. It is crucial to the variance-covariance method. The quality of approximation is best if we are measuring risk over a short time horizon and if portfolio value is almost linear in risk factor changes.

Stock Portfolio Example

Here there is no explicit time dependence in the mapping (4). The partial derivatives with respect to risk factors are

$$f_{z_i}(t, \mathbf{z}_t) = \alpha_i e^{z_{t,i}}, 1 \leq i \leq d,$$

and hence the linearized loss (6) is

$$L^{\Delta}_{t+1} = -\sum_{i=1}^{d} \alpha_i e^{\mathbf{z}_{t,i}} X_{t+1,i} = -V_t \sum_{i=1}^{d} \omega_{t,i} X_{t+1,i},$$

where $\omega_{t,i} = \alpha_i S_{t,i}/V_t$ is relative weight of stock i at time t. This formula may be compared with (5).

Example: European Call Option

Consider portfolio consisting of one standard European call on a non-dividend paying stock S with maturity T and exercise price K.

The Black-Scholes value of this asset at time t is $C^{BS}(t, S_t; r, \sigma)$ where $C^{BS}(t, S; r, \sigma) = S\Phi(d_1) - Ke^{r(T-t)}\Phi(d_2)$, Φ is standard normal df, r represents risk-free interest rate, σ the volatility of underlying stock, and where

$$d_1 = \frac{\log(S/K) + (r + \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} \text{ and } d_2 = d_1 - \sigma\sqrt{T-t}$$

. While in BS model, it is assumed that interest rates and volatilities are constant, in reality they tend to fluctuate over time; they should be added to our set of risk factors.

The Issue of Time

Scale Rather than measuring time in units of the time horizon (as we have implicitly done in most of this chapter) it is more common when derivatives are involved to measure time in years (as in the Black Scholes formula).

If Δ is the length of the time horizon measured in years (i.e. $\Delta = 1/260$ if time horizon is one day) then we have

$$V_t = f(t, \mathbf{Z}_t) = C^{BS}(t\Delta, S_t; r_t, \sigma_t).$$

When linearizing we have to recall that

$$f_t(t, \mathbf{Z}_t) = C_t^{BS}(t\Delta, S_t; r_t, \sigma_t)\Delta.$$

Example Summarized

The risk factors: $\mathbf{Z}_t = (\log S_t, r_t, \sigma_t)'$.

The risk factor changes: $\mathbf{X}_t = (\log(S_t/S_{t-1}), r_t - r_{t-1}, \sigma_t - \sigma_{t-1})'$.

The mapping (1)

$$V_t = f(t, \mathbf{Z}_t) = C^{BS}(t\Delta, S_t; r_t, \sigma_t),$$

The loss/loss operator could be calculated from (2). For derivative positions it is quite common to calculate linearised loss. The linearized loss (6)

$$L_{t+1}^{\Delta} = -\left(f_t(t, \mathbf{Z}_t) + \sum_{i=1}^{3} f_{z_i}(t, \mathbf{Z}_t) X_{t+1,i} \right).$$

The Greeks

It is more common to write the linearized loss as

$$L_{t+1}^{\Delta} = -(C_t^{BS}\Delta + C_S^{BS} S_t X_{t+1,1} + C_r^{BS} X_{t+1,2} + C_\sigma^{BS} X_{t+1,3}),$$

in terms of the derivatives of the BS formula.

- C_S^{BS} is known as the delta of the option.
- C_σ^{BS} is the vega.
- C_r^{BS} is the rho.
- C_t^{BS} is the theta.

Section 2: Risk Measures

Risk measures are used for the following purposes:

- Determination of risk capital. Risk measure gives amount of capital needed as a buffer against (unexpected) future losses to satisfy a regulator.
- Management tool. Risk measures are used in internal limit systems.
- Insurance premia can be viewed as measure of riskiness of insured claims.

Risk measure gives amount of capital that needs to be added to a position with loss L, so that the position becomes acceptable to an (internal/external) regulator.

Risk Measures based on Loss Distributions

The most popular risk measures like VaR describe the right tail of the loss distribution of L (or the left tail of the P&L).

To address this question we put aside the question of whether to look at conditional or unconditional loss distribution and assume that this has been decided.

Denote the distribution function of the loss $L := L_{t+1}$ by F_L so that $P(L \leq x) = F_L(x)$.

VaR and Expected Shortfall

- Primary risk measure: Value at Risk defined as

$$\mathrm{VaR}_\alpha = q_\alpha(F_L) = F_L^{\leftarrow}(\alpha) \tag{8}$$

i.e. the α-quantile of F_L.

- Alternative risk measure: Expected shortfall defined as

$$ES_\alpha = E(L|L \geq \mathrm{VaR}_\alpha) = \frac{1}{1-\alpha}\int_\alpha^1 q_u(F_L)du \tag{9}$$

i.e. the average loss when VaR is exceeded. ES_α gives information about frequency and size of large losses.

知道损失超过一定金额后损失的分布

VaR - badly defined!

The VaR bible is the book by Philippe Jorion.[Jorion, 2001].

The following "definition" is very common:

"VaR is the maximum expected loss of a portfolio over a given time horizon with a certain confidence level."

It is however mathematically meaningless and potentially misleading. In no sense is VaR a maximum loss!

We can lose more, sometimes much more, depending on the heaviness of the tail of the loss distribution.

Expected Shortfall

For continuous loss distributions expected shortfall is the expected loss, given that the VaR is exceeded.

For any $\alpha \in (0,1)$ we have

$$ES_\alpha = \frac{E(L; L \geq q_\alpha(L))}{1-\alpha} = E(L|L \geq \mathrm{VaR}_\alpha)$$

where we have used the notation $E(X; A) := E(XI_A)$ for a generic integrable rv X and a generic set $A \in \mathscr{F}$.

For a discontinuous loss df we have the more complicated expression

$$ES_\alpha = \frac{1}{1-\alpha}\left(E(L; L \geq q_\alpha(L)) + q_\alpha(1-\alpha-P(L \geq q_\alpha))\right)$$

Coherent Measures of Risk

There are many possible measures of the risk in a portfolio such as VaR, ES or stress losses. To decide which are reasonable risk measures a systematic approach is called for.

New approach of Artzner et.al.(1999):

- Give a list of properties (Axioms) that a reasonable risk measure should have; such risk measures are called coherent.
- Study coherence of standard risk measures (VaR, ES, stress losses etc.).
- On a more theoretical level: characterize all coherent risk measures.

The Axioms of Coherence

A coherent risk measure is a real-valued function ρ on some space of rv's (representing losses) that fulfills the following 4 axioms:

1. **Monotonicity**. For two rv's with $L_1 \geq L_2$ we have $\rho(L_1) \geq \rho(L_2)$.
2. **Subadditivity**. For any L_1, L_2 we have $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$. VaR不满足次可加性

   This is the most debated property. Necessary for following reasons:

   - Reflects idea that risk can be reduced by diversification and that "a merger creates no extra risk".
   - Makes decentralized risk management possible.

The Axioms of Coherence II

3. **Positive homogeneity**. For $\lambda \geq 0$ we have that $\rho(\lambda L) = \lambda \rho(L)$. If there is no diversification we should have equality in subadditivity axiom.

4. **Translation invariance**. For any $a \in \mathbb{R}$ we have that $\rho(L + a) = \rho(L) + a$.

金融资产整体上是一个偏度不大的分布；而巨灾保险一般不发生，一旦发生了就会产生很大亏损。
所以，金融上考虑VaR就足够了，保险领域更关注ES

Remarks:

- VaR is in general not coherent. ES (as we have defined it) is coherent.
- Non-subadditivity of VaR is relevant in presence of skewed loss distributions (credit-risk management, derivative books), or if traders optimize against VaR.

Section 3: Standard Statistical Methods for Market Risk

1. Variance-Covariance Method
2. Historical Simulation Method
3. Monte Carlo Simulation Method

Variance-Covariance Method

- We assume X_{t+1} has a multivariate normal distribution (either unconditionally or conditionally).
- We assume that the linearized loss in terms of risk factors is a sufficiently accurate approximation of the loss. We consider the problem of estimating the distribution of

$$L^{\Delta} = l^{\Delta}_{[t]}(\mathbf{X}_{t+1})$$

Theory Behind Method

Assume $\mathbf{X}_{t+1} \sim N_d(\mu, \Sigma)$.

Assume the linearized loss operator (7) has been determined and write this for convenience as

$$l^{\Delta}_{[t]}(\mathbf{x}) = -\left(c + \sum_{i=1}^{d} \omega_i x_i\right) = -(c + \mathbf{w}'\mathbf{x}).$$

The loss distribution is approximated by the distribution of $L^{\Delta} = l^{\Delta}_{[t]}(\mathbf{X}_{t+1})$.

Now since $\mathbf{X}_{t+1} \sim N_d(\mu, \Sigma) \Rightarrow \mathbf{w}'\mathbf{X}_{t+1} \sim N(\mathbf{w}'\mu, \mathbf{w}'\Sigma\mathbf{w})$, we have

$$L^{\Delta} \sim N(-c - \mathbf{w}'\mu, \mathbf{w}'\Sigma\mathbf{w}).$$

Implementing the Method

1. The constant terms in c and $\mathbf{w}$ are calculated.

2. The mean vector μ and covariance matrix Σ are estimated from data $\mathbf{X}_{t-n+1}, \cdots, \mathbf{X}_t$ to give estimates $\widehat{\mu}$ and $\widehat{\Sigma}$.

假设为正态分布后，只要估计2个参数，就能得到整个分布的信息

3. Inference about the loss distribution is made using distribution $N(-c-\mathbf{w}'\mu, \mathbf{w}'\Sigma\mathbf{w})$.

4. Estimates of the risk measures VaR_α and ES_α are calculated from the estimated distribution of L^Δ.

Estimating Risk Measures

- Value-at-Risk. VaR_α is estimated by

$$\widehat{\mathrm{VaR}}_\alpha = -c - \mathbf{w}'\widehat{\mu} + \sqrt{\mathbf{w}'\widehat{\Sigma}\mathbf{w})} \cdot \Phi^{-1}(\alpha)$$

- Expected Shortfall. ES_α is estimated by

$$\widehat{ES}_\alpha = -c - \mathbf{w}'\widehat{\mu} + \sqrt{\mathbf{w}'\widehat{\Sigma}\mathbf{w}} \cdot \frac{\phi(\Phi^{-1}(\alpha))}{1-\alpha}$$

Remark. For a rv $Y \sim N(0,1)$ it can be shown that $E(Y|Y > \Phi^{-1}(\alpha)) = \phi(\Phi^{-1}(\alpha))/(1-\alpha)$ where ϕ is standard normal density and Φ the df.

Pros and Cons, Extensions

- **Pros.** In contrast to the methods that follow, variance-covariance offers analytical solution with no simulation.

- **Cons.** Linearization may be crude approximation. Assumption of normality may seriously underestimate tail of loss distribution.

- **Extensions.** Instead of assuming normal risk factors, the method could be easily adapted to use multivariate Student t risk factors or multivariate hyperbolic risk factors, without sacrificing tractability. (Method works for all elliptical distributions.)

Historical Simulation Method

The Idea

Instead of estimating the distribution of $L = l_{[t]}(\mathbf{X}_{t+1})$ under some explicit parametric model for $\mathbf{X}_{t+1}$, estimate distribution of the loss operator under **empirical distribution** of data $\mathbf{X}_{t-n+1}, \cdots, \mathbf{X}_t$.

The Method

1. Construct the historical simulation data

$$\{\tilde{L}_s = l_{[t]}(\mathbf{X}_s) : s = t-n+1, \cdots, t\} \tag{10}$$

2. Make inference about loss distribution and risk measures using these historically simulated data: $\tilde{L}_{t-n+1}, \cdots, \tilde{L}_t$.

Inference about loss distribution

There are various possibilities in a simulation approach:

- Use empirical quantile estimation to estimate the VaR directly from the simulated data. But what about precision?
- Fit a parametric univariate distribution to $\tilde{L}_{t-n+1}, \cdots, \tilde{L}_t$ and calculate risk measures from this distribution. But which distribution, and will it model the tail?
- Use the techniques of **extreme value theory** to estimate the tail of the loss distribution and related risk measures.

Theoretical Justification

If $\mathbf{X}_{t-n+1}, \cdots, \mathbf{X}_t$ are iid or more generally stationary, convergence of empirical distribution to true distribution is ensured by suitable version of law of large numbers.

Pros and Cons

Pros. Easy to implement. No statistical estimation of the distribution of $\mathbf{X}$ necessary.

Cons. It may be difficult to collect sufficient quantities of relevant, synchronized data for all risk factors. Historical data may not contain examples of extreme scenarios.

The Monte Carlo Method

Idea

We estimate the distribution of $L = l_{[t]}(\mathbf{X}_{t+1})$ under some explicit parametric model for $\mathbf{X}_{t+1}$.

In contrast to the variance-covariance approach we do not necessarily make the problem analytically tractable by linearizing the loss and making an assumption of normality for the risk factors.

Instead we make inference about L using Monte Carlo methods, which involves simulation of new risk factor data.

The Method

1. With the help of the historical risk factor data $X_{t-n+1}, \cdots, X_t$ calibrate a suitable statistical model for risk factor changes and simulate m new data $\tilde{X}_{t+1}^{(1)}, \cdots, \tilde{X}_{t+1}^{(m)}$ from this model.

2. Construct the Monte Carlo data

$$\{\tilde{L}_i = l_{[t]}(\tilde{\mathbf{X}}_{t+i}^{(i)}), i = 1, \cdots, m\}$$

3. Make inference about loss distribution and risk measures using the simulated data $\tilde{L}_1, \cdots, \tilde{L}_m$. We have similar possibilities as for historical simulation.

Pros and Cons

Pros. Very general. No restriction in our choice of distribution for $\mathbf{X}_{t+1}$.

Cons. Can be very time consuming if loss operator is difficult to evaluate, which depends on size and complexity of portfolio.

Note that MC approach does not address the problem of determining the distribution of $\mathbf{X}_{t+1}$.